PostgreSQL + Student Lookup

CSC 425 | Linux - Ubuntu 24.04 LTS | Beginner lab

Name: _____ Branch: _____

You will create a database, add three sample students, and connect it to the Student Lookup already on the Student Dashboard. You will build GET to read students and POST to add a new student. Work through this PDF from beginning to end.

1. Get main and create your branch

Where: VS Code. You need your existing course repository clone, Git, Node.js 22 or later, and npm. This is a first-time PostgreSQL setup. If PostgreSQL is already installed, ask the instructor to check it before Step 2. Do not delete an existing database.

1. Choose **File > Open Folder** and open your course repository. The Explorer must show both `client` and `server`. This folder is the **repository root**. Keep the repository in a local folder, outside OneDrive, iCloud Drive, and other cloud-sync folders.

2. Choose **Terminal > New Terminal**. Use this terminal for the shell commands in the guide.

3. Type `git status` and press Enter. If it lists unfinished changes, save and commit those on their existing branch first. If they are on `main`, ask for help. Do not discard them.

4. With a clean working tree, run these commands one at a time. Replace `firstname-lastname` with your own name, using lowercase letters and hyphens:

```
git switch main
git pull --ff-only origin main
git switch -c firstname-lastname/student-database
git branch --show-current
```

Check: The final command prints your new personal branch. Stay on that branch for this entire lab. Never commit the lab on `main`.

5. Run these commands from the same terminal:

```
npm install
npm run install:all
```

6. Open `.gitignore` in the repository root. Add this line at the bottom and save:

```
server/local-postgres/
```

7. In the Explorer, right-click `server`, choose **New Folder**, and name it `local-postgres`. Create another folder under `server` named `endpointExamples`. If a folder already exists, use it.

Check: You have a personal branch and both folders. The database's data will live in `server/local-postgres/data`. Git stores your code; it must not store these live database files or your `.env` password file.

2. Install PostgreSQL on Linux

This guide uses Ubuntu 24.04 LTS. These exact package commands are for that version. If you use a different Linux distribution, ask the instructor to confirm the environment before continuing.

Where: the VS Code terminal at the repository root. This is a first installation on Ubuntu 24.04. If PostgreSQL is already installed, have the instructor check its login and storage location first. Do not stop or replace an existing server by following the new-install steps.

1. Install PostgreSQL 16, the version provided for this Ubuntu release:

```
sudo apt update
sudo apt install postgresql-16 postgresql-client-16
```

Enter your **Linux account password** if sudo asks. Nothing appears while you type. Press Enter. If apt asks whether to continue, type Y and press Enter.

2. Ubuntu automatically creates a default server outside your project. Stop that newly installed default and set it to manual startup so it will not occupy this lab's port after a reboot:

```
sudo pg_ctlcluster 16 main stop
echo manual | sudo tee /etc/postgresql/16/main/start.conf
```

3. Now create the lab's server inside your project. Run the next commands as your normal user, **without sudo**:

```
export PATH="/usr/lib/postgresql/16/bin:$PATH"
initdb -D server/local-postgres/data -U postgres -W -A scram-sha-256
```

4. At the two password prompts, type the same new database password and press Enter. No characters appear while you type. This password belongs to the database login **postgres**; remember it for the rest of the lab.

5. After initialization finishes, paste this entire block, including the backslash at the end of its first line:

```
pg_ctl -D server/local-postgres/data \
  -l server/local-postgres/server.log -o "-p 5432 -k /tmp" start
```

Check: The terminal says `server started`. If it reports a port conflict or another error, stop and ask for help. Do not run initdb again or delete the data folder.

6. Open the SQL console in this same terminal:

```
psql -h localhost -p 5432 -U postgres -d postgres
```

Enter your new database password. You should see `postgres=#`. Continue to Step 3 in that prompt. Do not type the prompt itself.

References: Ubuntu PostgreSQL installation, cluster startup settings.

3. Create the database and sample students

Where: psql, at the `postgres=#` **prompt.** Run these two lines separately:

```
CREATE DATABASE student_lookup;
\connect student_lookup postgres localhost 5432
```

Enter your database password if asked. The prompt should begin with `student_lookup`. Now check the actual connection:

```
\conninfo
```

Check: It names database **student_lookup**, user **postgres**, host **localhost**, and port **5432**. These are the four values you will put in `.env`. SQL ends with a semicolon; commands beginning with `\` do not. If the database already exists, have the instructor check it; do not delete it.

1. In VS Code, create `server/endpointExamples/setup.sql`. Paste all of the following into the file and save it:

```
CREATE TABLE students (
  student_id INTEGER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  first_name TEXT NOT NULL,
  last_name TEXT NOT NULL,
  major TEXT NOT NULL
);

INSERT INTO students (first_name, last_name, major)
VALUES
  ('Avery', 'Morgan', 'Computer Science'),
  ('Jordan', 'Bennett', 'Information Systems'),
  ('Taylor', 'Reed', 'Business Administration');

SELECT * FROM students ORDER BY student_id;
```

2. Select all text in the file with **Ctrl+A**, then copy with **Ctrl+C**. Click inside the psql window and **press Ctrl+Shift+V**. Press Enter. Run this setup script **once**, not every time you start the app.

Check: The result shows three rows: Avery Morgan (ID 1), Jordan Bennett (ID 2), and Taylor Reed (ID 3). If a result viewer shows `(END)`, press `q`. Leave psql by typing `\q` and pressing Enter.

4. Save the connection settings

Where: VS Code Explorer. Create a file called `.env` directly inside `server`. If it exists, update the entries below. Do not create `.env.txt` or put the file in `client`.

```
PORT=3000
DB_HOST=localhost
DB_PORT=5432
DB_NAME=student_lookup
DB_USER=postgres
DB_PASSWORD="YOUR_DATABASE_PASSWORD"
```

Replace `YOUR_DATABASE_PASSWORD` with the password that just worked in psql; keep the quotes. Save the file. This is your PostgreSQL password, not your computer or GitHub password. Keep `.env` out of Git and screenshots.

5. Create the database connection and queries

Where: VS Code. Create `server/endpointExamples/studentDb.js`. Paste this entire code block into it, then save:

```
import dotenv from 'dotenv'
import pg from 'pg'

dotenv.config({
  path: new URL('../.env', import.meta.url),
  quiet: true,
})

const pool = new pg.Pool({
  host: process.env.DB_HOST,
  port: Number(process.env.DB_PORT),
  database: process.env.DB_NAME,
  user: process.env.DB_USER,
  password: process.env.DB_PASSWORD,
  connectionTimeoutMillis: 3000,
})

export async function getStudents() {
  const result = await pool.query(`
    SELECT student_id::text AS "studentID",
           first_name AS "firstName", last_name AS "lastName", major
    FROM students
    ORDER BY student_id
  `)
  return result.rows
}

export async function addStudent(firstName, lastName, major) {
  const result = await pool.query(`
    INSERT INTO students (first_name, last_name, major)
    VALUES ($1, $2, $3)
    RETURNING student_id::text AS "studentID",
              first_name AS "firstName", last_name AS "lastName", major
  `, [firstName, lastName, major])
  return result.rows[0]
}
```

What this file does: `dotenv` reads your private settings. The pool connects to PostgreSQL. `getStudents()` reads rows. `addStudent()` inserts a row and returns it. PostgreSQL assigns the new ID.

`await` waits for the database. $1, $2, and $3 receive the values supplied separately in the array. The field names `studentID`, `firstName`, `lastName`, and `major` match the existing Student Lookup. This file handles data; the next file handles HTTP requests.

Check: The filename is `studentDb.js`, with a capital D. It is saved inside `server/endpointExamples`. You do not run this file by itself. Query reference

6. Create the Express GET and POST endpoints

Where: VS Code. Create `server/endpointExamples/studentServer.js`. Paste this entire block into it and save:

```
import express from 'express'
import { getStudents, addStudent } from './studentDb.js'

export const studentRoutes = express.Router()

studentRoutes.get('/', async (req, res) => {
  const students = await getStudents()
  res.json({ students })
})

studentRoutes.post('/', async (req, res) => {
  const { firstName, lastName, major } = req.body ?? {}
  if (
    typeof firstName !== 'string' || firstName.trim() === '' ||
    typeof lastName !== 'string' || lastName.trim() === '' ||
    typeof major !== 'string' || major.trim() === ''
  ) {
    return res.status(400).json({ message: 'Complete all three fields.' })
  }

  const student = await addStudent(
    firstName.trim(), lastName.trim(), major.trim(),
  )
  res.status(201).json({ message: 'Student added.', student })
})

studentRoutes.use((error, req, res, next) => {
  console.error(error.message)
  res.status(500).json({ message: 'Database request failed.' })
})
```

GET calls `getStudents()` and returns `{ "students": [...] }` as JSON. This is the response your frontend already expects.

POST reads three fields from `req.body`, checks them, calls `addStudent()`, and returns the saved student with status **201 Created**. Missing or blank fields return **400**.

The final handler sends a JSON error if a database query fails. Express 5 forwards rejected async handlers to it. The SQL remains in `studentDb.js`, so you can see the difference between the database code and the API code.

Check: This file has one GET handler and one POST handler. It exports `studentRoutes`. The `/` paths will become `/api/students` when you attach this router in Step 7. Do not add `app.listen()` here. Express router reference

7. Attach the routes to the existing server

Where: `server/src/app.js`**.** Make these four edits carefully. Do not replace the entire file; the transcript exercise also uses it.

1. At the top, below the existing import lines, add:

```
import { studentRoutes } from '../endpointExamples/studentServer.js'
```

2. Find the old `teachingApi.get('/students', ...)` handler. Delete that entire handler, including its comments. In the current main branch, it looks like this:

```
teachingApi.get('/students', (_request, response) => {
  // Express converts the JavaScript object to JSON for the React frontend.
  // StudentLookup selects the matching full name from this returned array.
  response.json({ students: mockStudents })
})
```

This is the old hard-coded response. Removing it lets requests reach your new database routes. Leave the `mockStudents` array for now; the older `/api/status` route still uses its count.

3. Find the line that starts `teachingApi.use(['/status', '/students', '/transcripts']`. Remove only `'/students',` from its list. The finished statement should be:

```
teachingApi.use(
  ['/status', '/transcripts'],
  cors({ methods: ['GET', 'HEAD', 'OPTIONS'] }),
)
```

4. Lower in the file, find `app.use(express.json())`. Add your router immediately after it. These three lines should now appear together, in this order:

```
app.use(cors({ origin: 'http://localhost:5173' }))
app.use(express.json())
app.use('/api/students', studentRoutes)
```

Save the file. Do not duplicate the existing `cors` or `express.json` lines. `express.json()` reads POST request bodies. The last line connects `/api/students` to the handlers you wrote.

Check: Searching `app.js` for `teachingApi.get('/students'` finds nothing. Searching for `app.use('/api/students', studentRoutes)` finds exactly one match. `server/src/index.js` already starts this Express app on port 3000; no extra server is needed.

8. Run the app and check GET

Where: an ordinary VS Code terminal at the repository root. If you still see a psql prompt, type `\q` and press Enter first.

1. Run:

```
npm run dev
```

Leave this terminal running. It starts Express and the frontend together. **Check:** The API reports `http://localhost:3000`, and Vite reports `http://localhost:5173`. If a port is occupied, stop the earlier copy you started, then retry. Ask for help if you cannot identify it.

2. In your browser, open `http://localhost:3000/api/health`. Confirm it says `"database":"connected"`. Then open `http://localhost:3000/api/students`.

Check: You see JSON containing a `students` array with your three database students. Avery's ID is the string `"1"`. If you see five students with IDs 1001-1005, Step 7 still has the old mock handler. If you see `Database request failed.`, confirm PostgreSQL is running and check `server/.env`; restart the API after editing `.env`.

3. Open `http://localhost:5173/#student-lookup`. Enter **Avery Morgan** in **Student full name**, then select **Search**. You should see ID 1 and Computer Science.

9. POST a new student, then search again

1. Open a **second VS Code terminal** at the repository root. Type `node` and press Enter. You should see a > prompt. The following block is JavaScript for this Node prompt, not a shell command.

2. Paste the whole block and press Enter. Run it once:

```
var response = await fetch('http://localhost:3000/api/students', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    firstName: 'Morgan',
    lastName: 'Ellis',
    major: 'Computer Science',
  }),
})
var result = await response.json()
console.log(response.status, result)
```

Check: You see **201**, `Student added.`, and the saved student's ID. A fresh database normally assigns ID 4. Use the returned value if yours differs.

3. In Student Lookup, search for **Morgan Ellis**. Confirm the displayed ID matches the POST response. The React form already calls GET `/api/students`; you do not need to rewrite it. New directory students do not automatically receive a transcript in the separate transcript card.

4. Take a screenshot of the successful lookup. Type `.exit` at the Node prompt when you finish. If you accidentally POST the same name twice, ask the instructor for help; the lookup reports duplicate names.

10. Save your work and come back later

Where: VS Code. Create `server/endpointExamples/lab-notes-firstname-lastname.md`, using your own name in the filename. Write your branch name, the ID returned by POST, and the POST status code. Add a short explanation of this sequence:

```
Student Lookup -> GET /api/students -> studentServer.js
               -> studentDb.js -> PostgreSQL
               -> rows -> JSON response -> Student Lookup
```

Save your screenshot as `server/endpointExamples/lookup-firstname-lastname.png`. Keep passwords out of it. Then run the following commands in an ordinary terminal at the repository root. Replace `firstname-lastname` with your name:

```
git status
git add .gitignore server/src/app.js
git add server/endpointExamples/setup.sql
git add server/endpointExamples/studentDb.js
git add server/endpointExamples/studentServer.js
git add server/endpointExamples/lab-notes-firstname-lastname.md
git add server/endpointExamples/lookup-firstname-lastname.png
git diff --cached --stat
git commit -m "Connect Student Lookup to PostgreSQL"
git push -u origin firstname-lastname/student-database
```

Check before committing: Only your code, SQL, notes, screenshot, and `.gitignore` are staged. `.env` and `server/local-postgres/data` must not appear. Submit your GitHub branch link and screenshot in the course assignment. Do not merge into `main`.

When you finish today

Press **Ctrl+C** in the terminal running `npm run dev`. In an ordinary terminal at the repository root, run:

```
/usr/lib/postgresql/16/bin/pg_ctl \
  -D server/local-postgres/data -m fast stop
```

When you return to the project

At the repository root, run this block once to start the stopped database:

```
export PATH="/usr/lib/postgresql/16/bin:$PATH"
pg_ctl -D server/local-postgres/data \
  -l server/local-postgres/server.log -o "-p 5432 -k /tmp" start
```

Then run `npm run dev` from the repository root and open `http://localhost:5173/#student-lookup`. Search for Morgan Ellis again. The record remains in PostgreSQL after the API stops.

Do not repeat initialization, CREATE DATABASE, or the seed INSERT when restarting. Keep your project in the same folder. Changing Git branches does not reset the database, and pushing code does not upload your local database records.